ALEXANDRA ABÓS

Lead Data Scientist | AI & Data Science | Life Sciences

Barcelona, Spain | LinkedIn | GitHub | Google Scholar

PROFILE

Lead Data Scientist with a PhD in Biomedical Engineering / Medicine and Translational Sciences, combining machine learning, statistical modelling and technical leadership to build scalable AI solutions at the intersection of data, business and life sciences. Currently leading the technical direction of a global AI initiative at Sanofi, with solutions deployed across 20+ countries and multiple products. Experienced in technical stakeholder management, mentoring Data Scientists, Data Science standardization and operationalization, and translating business problems into scalable Data Science solutions.

PROFESSIONAL EXPERIENCE

Sanofi — Lead Data Scientist

Sep 2024 – Present | Barcelona, Spain

- Lead the design and development of an omnichannel call plan optimization system, deployed across 20+ countries and multiple products to support commercial engagement decisions.
- Act as the technical go-to person for business and cross-functional stakeholders, translating business needs into scalable AI and Data Science solutions.
- Mentor 3 internal Data Scientists and provide technical guidance to a wider group of 6 external Data Scientists, supporting technical development and delivery.
- Define methodologies, standards and best practices together with other Data Science Leads to improve consistency, scalability and quality of Data Science solutions.
- Drive the standardization and evolution of the Data Science pipeline, improving usability and enabling more self-service capabilities for Data Scientists and users.
- Collaborate with Data Engineering, MLOps, business and other technical stakeholders to scale and operationalize Data Science solutions.

Selected projects: OCCP optimization · Data Science pipeline standardization · Pharmacokinetic modelling

Sanofi — Data Scientist

May 2022 – Aug 2024 | Barcelona, Spain

- Developed machine learning and advanced analytics solutions supporting commercial and R&D decision-making across multiple business areas.
- Designed optimization algorithms for omnichannel engagement and recommendation systems.
- Built user profiling and segmentation approaches for personalized recommendations and data-driven targeting.
- Developed forecasting models for seasonal demand at sub-national level, enabling optimization of digital media campaigns.
- Assessed the short- and mid-term impact of COVID-19 across brands and therapeutic areas to support business planning.
- Developed statistical and machine learning approaches to improve pharmacokinetic modelling in drug discovery, contributing to research initiatives and scientific publications.

Technologies: Python, PySpark, MLflow, PyTorch, AWS, Databricks, Docker, Git, CI/CD

Sanofi — Statistical Forecaster

Jul 2019 – Jun 2022 | Barcelona Metropolitan Area

- Managed monthly demand forecasting cycles across North America, Europe, LATAM, JPAC and China.
- Developed, evaluated and maintained forecasting models ranging from statistical approaches to machine learning, including MLR, ETS, ARIMAX and Elastic Net.
- Conducted root-cause analysis of forecast errors and identified opportunities to improve model accuracy and prioritize deployment across markets and portfolios.
- Supported the implementation of forecasting processes and tools as part of a broader Supply Chain transformation programme.
- Collaborated with Supply Chain Business Partners to incorporate market events and business knowledge into forecasting models and consensus planning.

UPF Barcelona School of Management — Programme Coordinator & Lecturer

Oct 2023 – Present | Barcelona, Spain

Programme coordination and teaching within the Master in Data Analytics for Business, including Python, exploratory data analysis, data visualization and health data analytics.

RESEARCH & ACADEMIC EXPERIENCE

Universitat de Barcelona — PhD / Early Stage Researcher

Oct 2015 – Jul 2019 | CJNeuroLab, Department of Medicine

Research focused on applying Data Science and statistical approaches to neuroimaging and brain connectivity, with the goal of improving understanding of network structure and integrity in neurological disorders.

External Collaborator, Aug 2019 – Aug 2022

Research highlights: 25+ scientific publications · 2 patents · Research experience at the University of Oxford

EDUCATION

PhD in Medicine and Translational Sciences — Universitat de Barcelona

Biomedical Engineering — Universitat de Barcelona (Bachelor's and Master's)

TECHNICAL EXPERTISE

AI & Machine Learning · Statistical Modelling · Optimization · Commercial Analytics · Pharmacokinetic Modelling · Forecasting · Python · PySpark / Spark · PyTorch · MLflow · AWS · Databricks · Docker · Git · CI/CD · Data Science Engineering · MLOps · Technical Leadership · Mentoring · Data Science Standardization

LANGUAGES

Spanish · Catalan · English · French · Japanese